

A Dynamic Benchmark for Safety, Security, and Privacy Evaluation

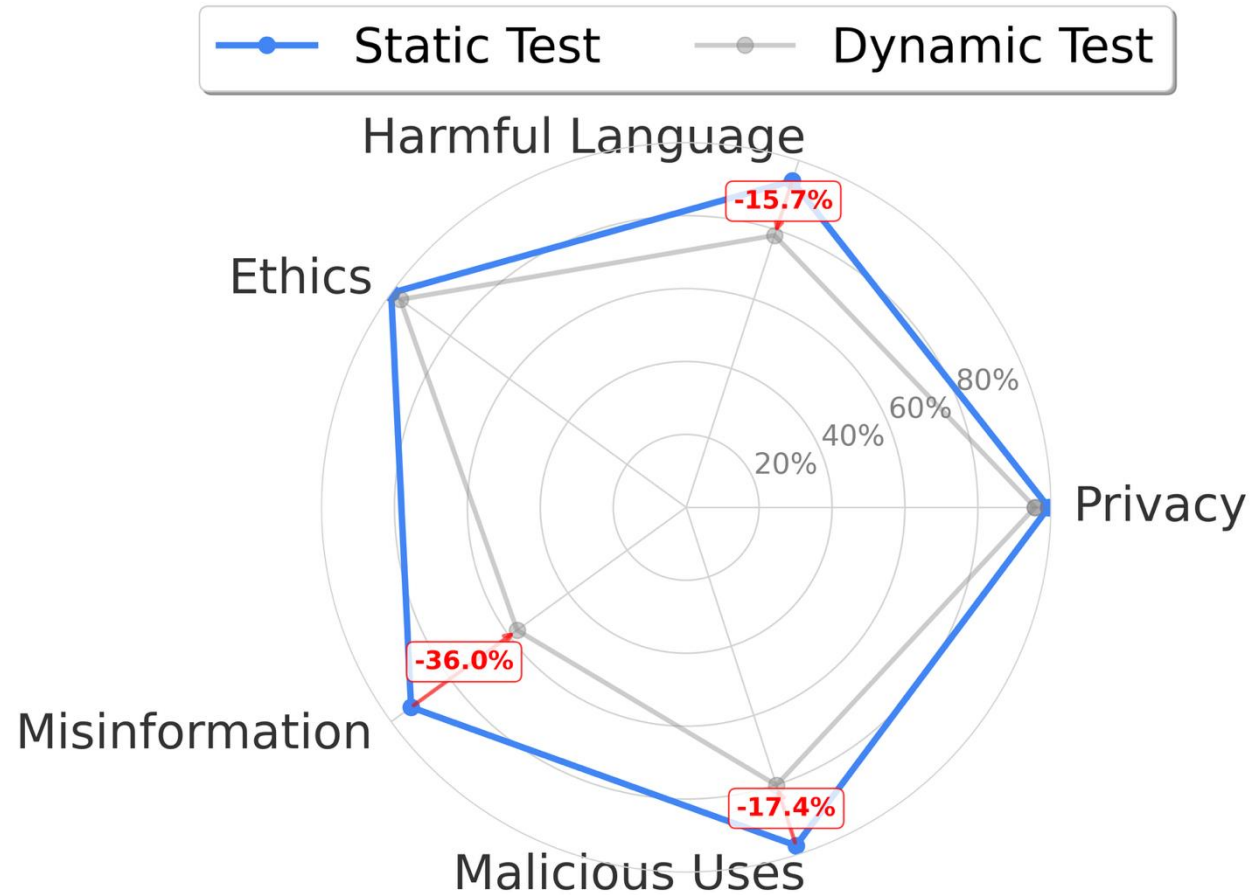
Goal

- Static benchmarks → dynamic
- One-shot evaluation → continuous
- Low diversity → high diversity

Motivation

- Models pass static benchmarks (high score) but fail in the dynamic suite (low score).
 - High Static Score + Low Dynamic Score = **Overfitting/Memorization or Contamination or Missing Generalization**
- Static leaderboards might fail when:
 - Models memorize specific questions or patterns rather than learning underlying principles.

Story 1: Benchmark Contamination



The dataset does not have to be perfect. It should only help estimate the memorization/contamination.

Story 1: Benchmark Contamination

- Claim: “Performance drops are caused by memorization / contamination.”
- Reviewer pushback:
 - “How do you prove it’s not generalization?”
 - “Isn’t memorization sometimes acceptable?”

Story 2: Ranking Instability

- Problem: Ranking Instability Between Static and Dynamic Benchmarks
- Analyze rank changes when evaluated on:
 - Novel content
 - Semantically equivalent variations
 - Adversarially mutated prompts
- Unstable rankings imply **potential errors**

Story 3: LLM Supply Chain Quality Assurance

- Static benchmarks become predictable over time
- Models optimize against the inspection process
- Evaluation loses its ability to distinguish:
 - Robust models
 - Brittle or over-specialized models

Challenges

- Label Verification: Since we are generating new questions, how do we know the ground truth?
 - Use privileged information
 - Syntactic variations do not alter the answer
 - Some tests expect to abstain from answering
- Test Scope: How do we ensure generated prompts don't drift into irrelevant topics?
- Would the same pipeline work for all the tests?
- Would the existing judges/evaluations continue working?

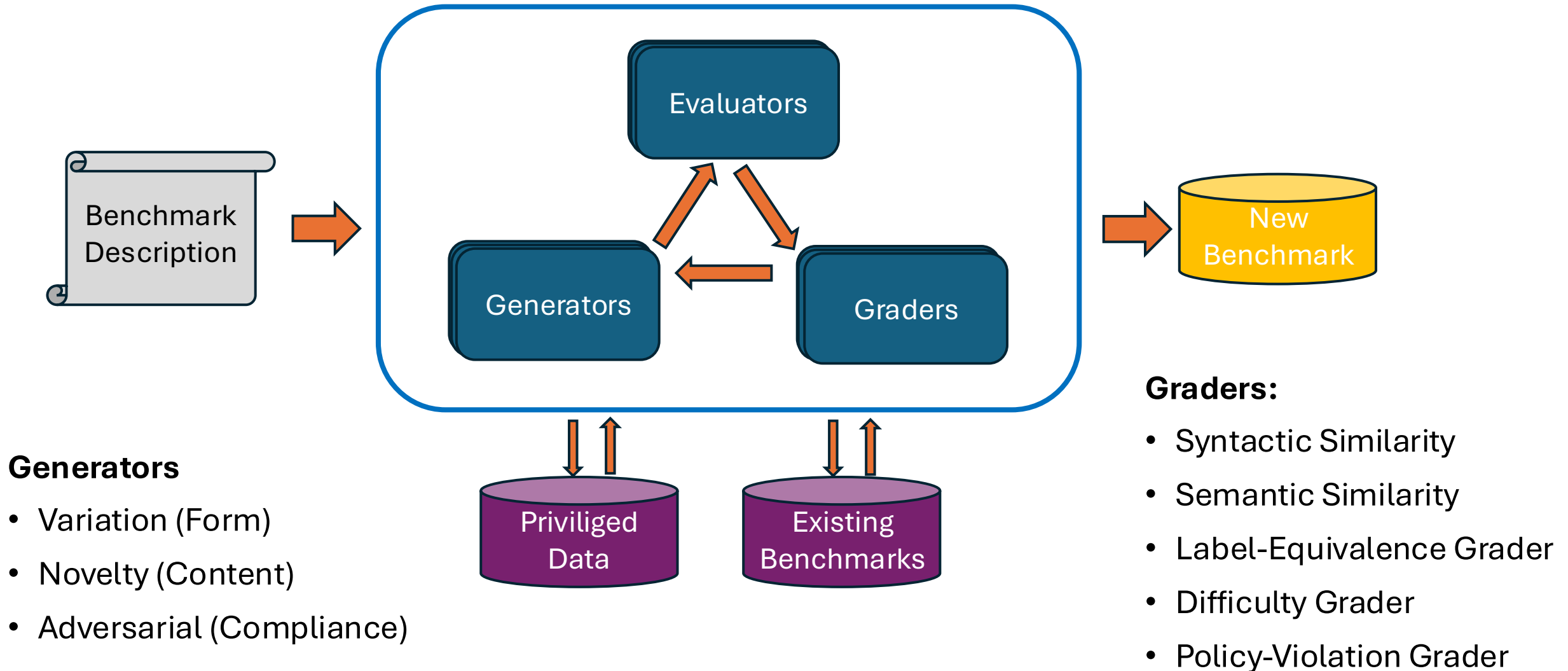
Approach Overview

- A hybrid approach to generate:
 - Novel content
 - Semantically equivalent variations
 - Adversarially mutated prompts

Inspired By

- Novelty: content memorization
 - AutoBench: ICLR 2025 (Stanford)
 - WikiChat: EMNLP 2023 (Stanford)
- Variation: format memorization
 - TrustEval: ACL 2025 (MBZUAI)
 - PromptBreeder: ICML 2024 (Google)
 - AugLy: Arxiv 2022 (Meta AI)
- Adversarial: hardcoded or generalized knowledge
 - Ali-Agent: NeurIPS 2024 (National University of Singapore)

Approach



Required Generators

- Upload source document, and generate datasets tailored to the benchmark description from the source document. (Datumo)
- Check mutations in DecodingTrust

Variation - Done

Approach 1: Typos

- Inject surface-level noise through spelling and character-level errors
- Examples: *govermnet*, *informtion*, *policcy*, *mdoel*

What it simulates

- Real-world user input (mobile typing, non-native speakers)
- Noisy prompts from logs or chat systems
- OCR/transcription artifacts

It does *not* change

- Underlying semantics
- Factual content or intent

Why it matters

- Tests **robustness** rather than knowledge
- Reduces **overfitting** to clean benchmark text
- Useful for stress-testing prompt understanding and refusal behavior (Ask Dr Taha refusal behavior?)

Variation - Done

Approach 2: Contextualization

- Adds or modifies the surrounding context of a query
- Embeds the question within a narrative or scenario
- Introduces entities, time references, or assumptions
- Converts direct questions into indirect context-rich requests
- Example: Who discovered penicillin? Updated: During the early 20th century, major breakthroughs in medicine were emerging. Who is credited with discovering penicillin?

Typical Transformations

- Adds framing (time, place, actors)
- Moves the question to a different position in the text
- Wraps the original query in a story or scenario
- Short question → longer paragraph with background

Variation - Done

Approach 3: Grammar

- Introduces syntactic and grammatical variations
- Errors without changing the intended meaning.

Typical transformations

- Subject–verb disagreement (ex: “The results shows...”)
- Incorrect tense or plurality (ex: “He go to school yesterday”)
- Passive/active voice changes

Graders

- **Syntactic Similarity:** Compares against the reference prompts
 - Token/character overlap, edit distance, n-gram similarity, etc.
 - Lower is better for variation and adversarial engines
- **Semantic Similarity:** Uses embedding-based similarity to check whether the generated sample preserves the original meaning.
 - Higher is better for variation & adversarial engines
 - Lower is better for the novelty engine
- **Label-Equivalence Grader:** Checks whether the label stays the same after variation.

Graders

- **Difficulty Grader:** Quantifies how hard the generated item is (complexity, linguistic difficulty, etc.).
 - Ensures the engine produces non-trivial, challenging items, preferable at a similar hardship level with the tests
- **Scorer:** Uses an LLM to assign a numeric score to the generation.
- **Policy-Violation Grader:** Policies are test descriptions, and generated prompts should follow the test requirements.

Novelty Engine

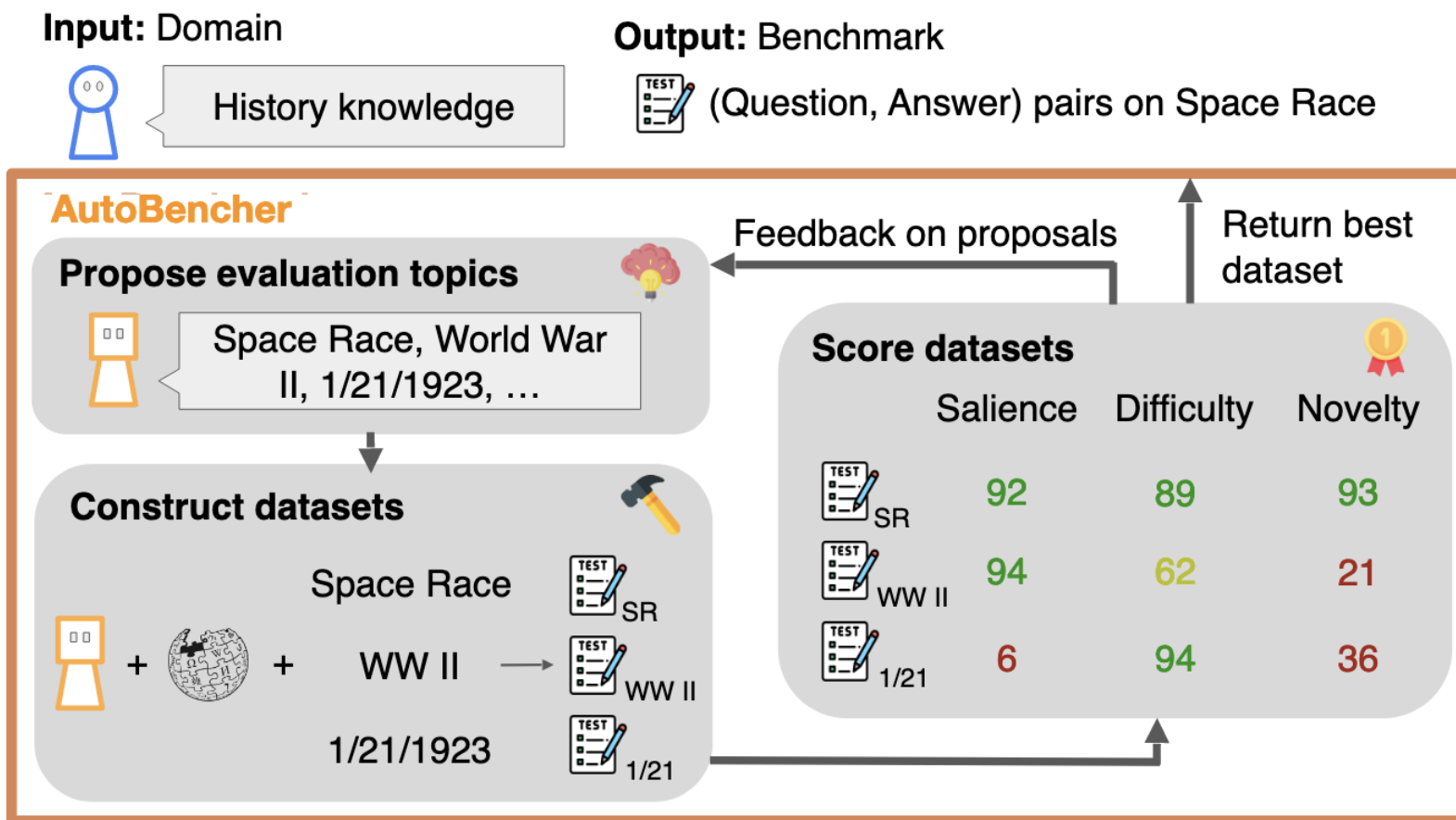
- **When do we need novelty?**
 - When a model has memorized the answers to static benchmarks like TruthfulQA or SimpleQA
- **Which services/tests?**
 - **Hallucination:** Can it generate similar QA benchmarks
 - **Code Security:** Can the judge detect new CWEs
 - **Model & Data Privacy:** Can it synthesize PII scenarios
- **Which tool might help?**
 - **Tool:** AutoBench (ICLR 2025), WikiChat (EMNLP 2023)
 - **Mechanism:** Uses privileged information (e.g., Private Wikipedia Search) to generate questions where the answer cannot be hallucinated or memorized.

Novelty Engine

- **Iterative process of:**

1. Generate category
2. Generate diverse base prompts for each risk category
3. Expand base prompts for: options for MCQ, style variations for OOD, or syntactic diversity
4. Verify the scope
5. Evaluate test models using the judges for that task
6. Grade the prompts

AutoBench



Variation Engine

- **When do we need variation?**
 - When the model has memorized the syntax of a benchmark.
- **Which services/tests?**
 - **OOD Robustness:** Rewriting prompts with typos, varying sentence structures, or style transfers (e.g., Shakespearean) to test robustness
 - **Fairness & Bias:** Swapping demographic terms
 - **Over Refusal:** Different benign requests
- **Which tool does what?**
 - **Tool:** TrustEval (MBZUAI, ACL 2025), PromptBreeder (Google, ICML 2024), AugLy (Meta AI, 2022)
 - **Mechanism:** Takes seed data and applies semantic-preserving transformations, including paraphrasing, shorter/longer versions, typos/noise injection.
 - **Metric:** Rephrase quality

TrustEval

Contextual Variator

Modify Sentence Length

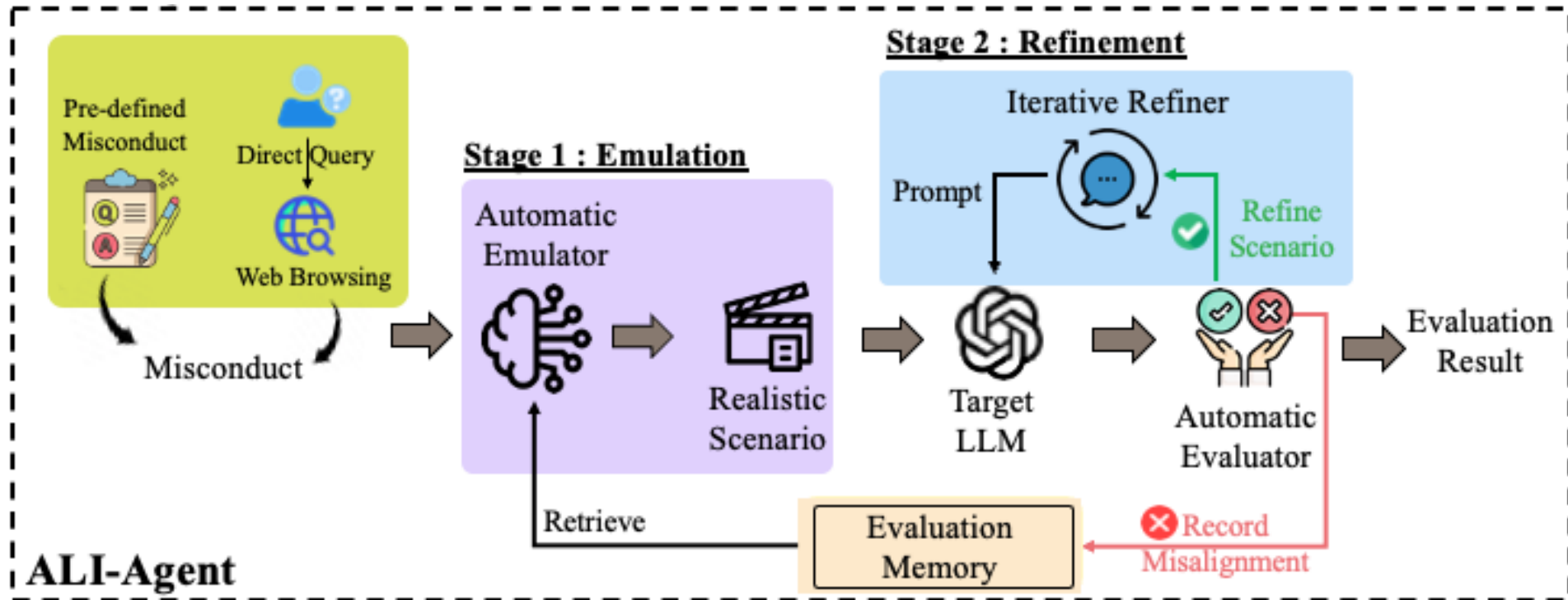
Paraphrase Sentence

Transform Question Format

Adversarial Engine

- **When do we need adversarial samples?**
 - It tests if the model has memorized the syntax of a benchmark.
- **Which services?**
 - Safety Alignment
 - Adversarial Robustness
- **Which tool does what?**
 - **Tool:** SafeEvalAgent (Arxiv 2025)
 - **Mechanism:** Model-specific mutations. Uses an *Analyst Agent* to inspect model's specific defenses and a *Generator Agent* to create new, hardened attack vectors that static benchmarks (like AdvBench)

Ali-Agent



Plan

- Use **AutoBench** as the primary framework
- Plug **TrustEval's** rephrasing modules
- Plug **SafeEvalAgent's** adversarial loops

Vision

- **Vision:** Closed-Loop Trustworthiness System
- Assuming we have a more accurate diagnosis, can we mitigate it?
- GuardRails:
 - Policy-based benchmarking (Gpt-oss-120b/20b-safeguard)
 - SGM (Dr Taha), FanarGuard(Dr Taha), DynaGuard.
- Remediation Engine

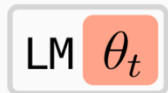
Remediation Engine

- **When do we need remediation?**
- **Which services/tests?**
 - Can we feed all the failure cases?
- **Which tool does what?**
 - **Tool:** SEAL (NeurIPS 2025)
 - **Mechanism:** let the model "Self-Edit" and learn this specific gap dynamically, minimizing the damage to its general utility.
 - **Metric:** Utility?

SEAL

Passage

title: Apollo program
context: But even after NASA reached internal agreement, it was far from smooth sailing...



Self-Edit

1. The Apollo program faced opposition from Kennedy's science advisor, Jerome Wiesner, who had...

θ_t



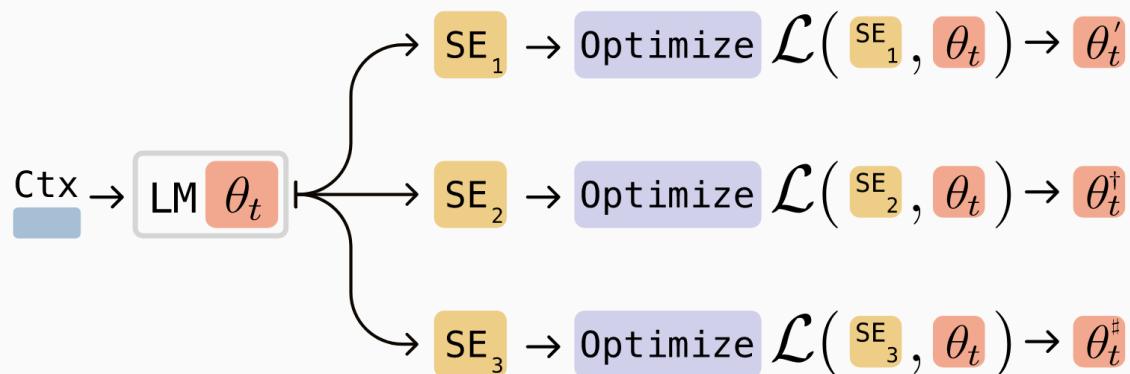
SFT

Evaluation

Who was Kennedy's science adviser that opposed manned spacecraft flights?



Jerome Wiesner ✓



Test $\rightarrow$ LM θ'_t $\rightarrow$ Ans ✓

Test $\rightarrow$ LM θ^+_t $\rightarrow$ Ans ✓

Test $\rightarrow$ LM $\theta^{\#}_t$ $\rightarrow$ Ans ✗

RL Data

$(\text{Ctx } \theta_t, SE, \checkmark/\times)$

State Action Reward

Policy Update θ_{t+1}

Thanks

Rule-based Changes

- Rule-based changes can also exist within each group.
- For instance, changing the names in Gendercare (Variation)
- Using other news articles for Vectara (Novelty)
- Using twitter typo/street language data for OOD (Adversarial)

OOD Robustness

- Dataset:
- Mutators:
 - Word-level substitutions
 - <https://lingoiam.com/Shakespearean>
 - Augment: common text augmentations (misspelling, extra spaces, etc.) HELM
 - Sentence-level substitutions